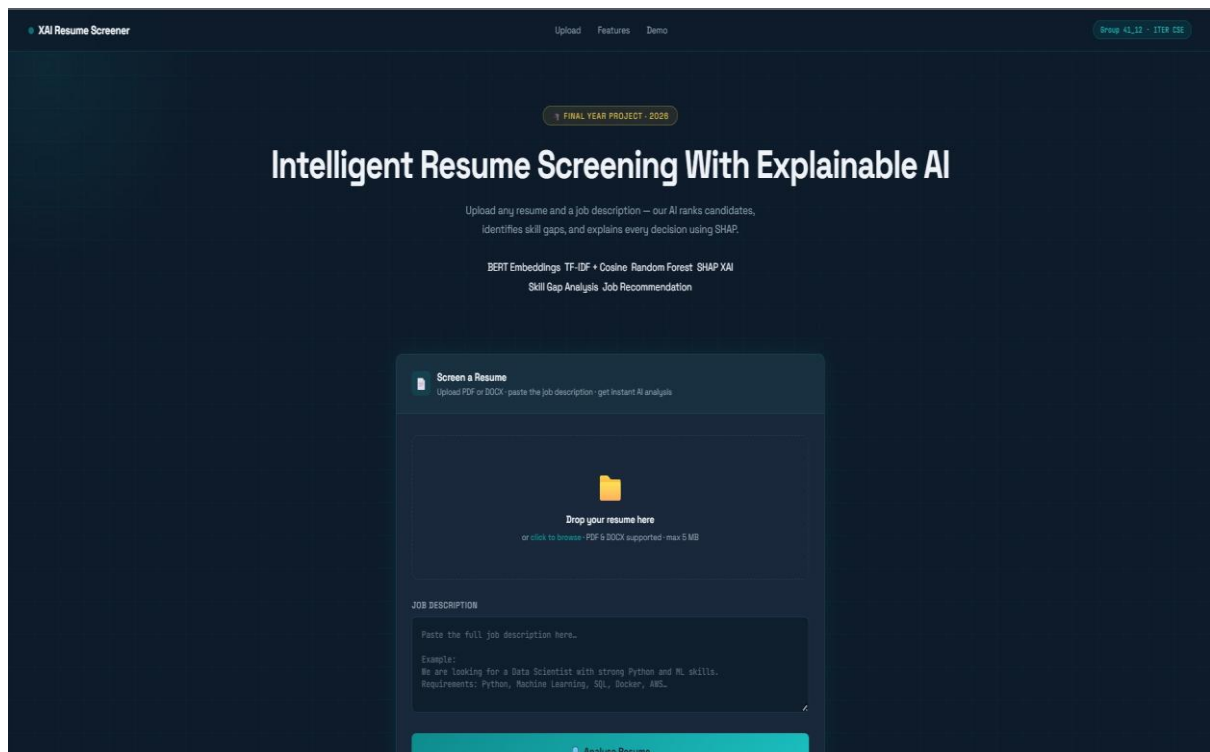


Resume Screening with Explainable AI - Project Screenshots & Results

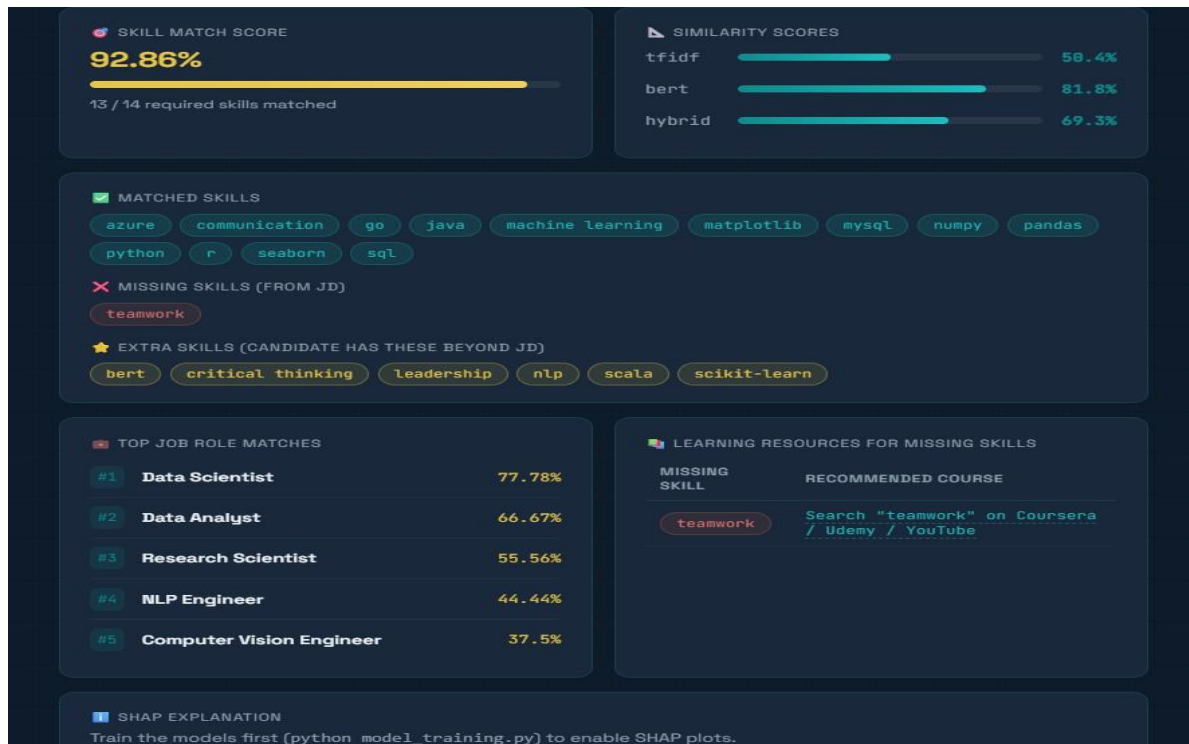
Piyush Kumar Nayak | B.Tech CSE | ITER, SOA University

Figure 1: Home Page / Resume Upload Interface



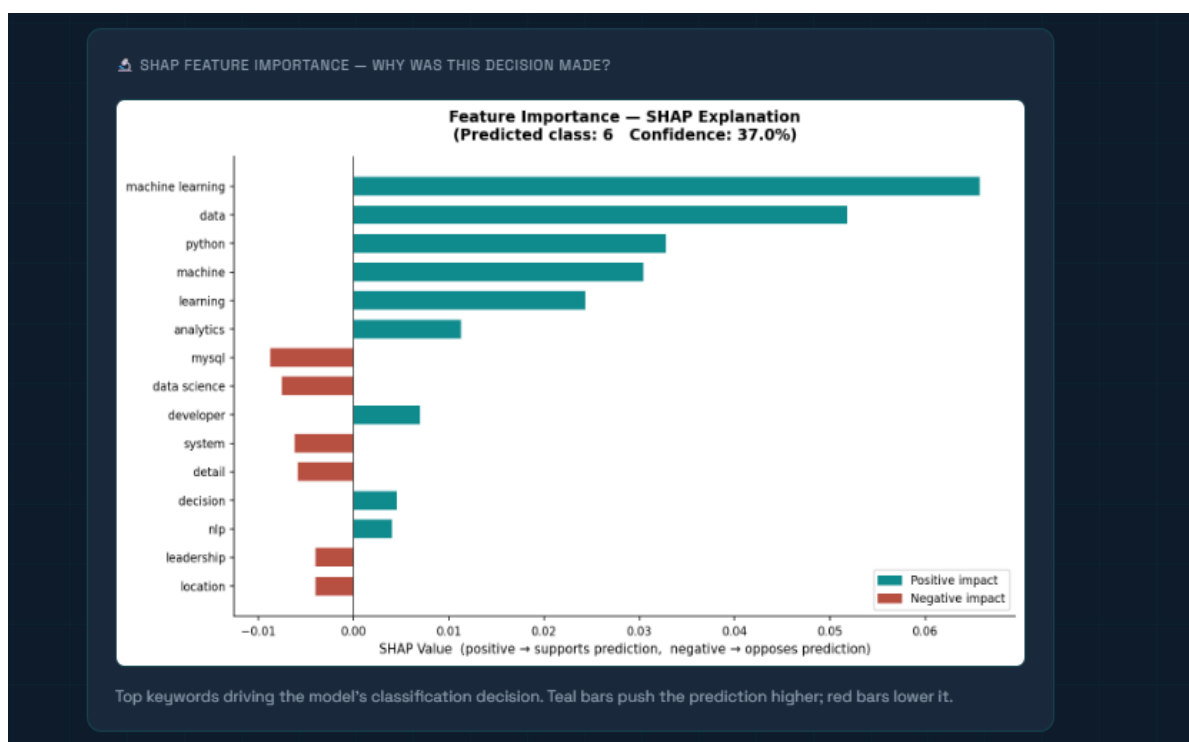
This is the main user interface of the application. Users upload a resume (PDF/DOCX) and paste a job description. The system uses TF-IDF, BERT embeddings, Random Forest classification, and SHAP explainability to analyze candidates.

Figure 2: Skill Gap Analysis and Job Recommendation Results



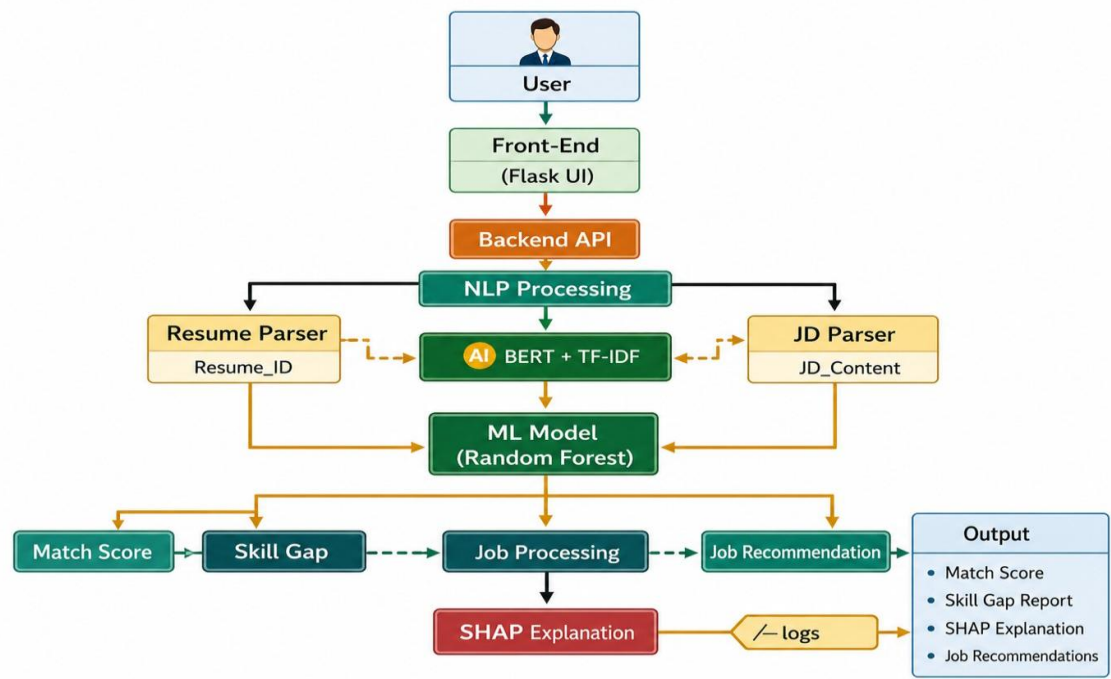
The system computes a skill match score of 92.86%, identifies matched and missing skills, recommends learning resources, and predicts the most suitable job roles. This demonstrates the effectiveness of the recommendation engine.

Figure 3: SHAP Explainability Results



SHAP feature importance visualization explains why a particular prediction was made. Positive features support the classification while negative features reduce confidence, making the AI model transparent and interpretable.

Figure 4: System Architecture Diagram



The architecture shows the complete workflow: Resume Parsing, Job Description Parsing, NLP Processing, TF-IDF and BERT feature extraction, Random Forest classification, Skill Gap Analysis, Job Recommendation, and SHAP Explainability. Outputs include match score, recommendations, and explanation reports.

Project Result Summary

The developed Explainable AI framework successfully performs intelligent resume screening, skill gap identification, candidate ranking, and job recommendation. The system integrates TF-IDF, BERT embeddings, Random Forest classification, and SHAP-based explainability to provide transparent and reliable hiring insights.